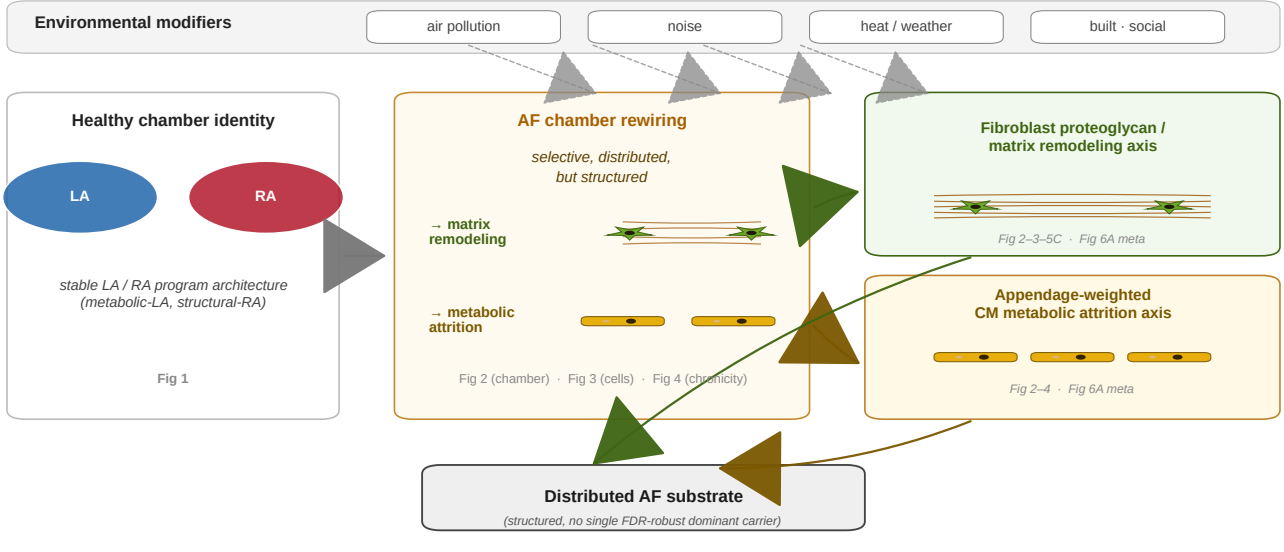

Substrate phenotyping beyond a binary AF / SR state

matrix–metabolic two-axis framework · cohort-stable structural arm · appendage-weighted metabolic decline